

Terasakiella magnetica sp. nov., *Magnetovibrio massiliensis* sp. nov. and *Magnetococcus organivorans* sp. nov., three novel magnetotactic bacteria isolated from the Mediterranean Sea

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Abstract

Three magnetotactic bacteria, designated strains PR-1^T, PR-2^T and PR-3^T, were isolated from seawater collected from the Mediterranean Sea at the Pointe Rouge Marina in Marseille, France. Phylogenetic analysis based on 16S rRNA gene sequences showed that strain PR-1^T formed a clade with representatives of the genus *Terasakiella*, which previously did not include any magnetotactic members. Strains PR-1^T shared 97.90% sequence similarity with *Terasakiella brassicae* B3^T. Strains PR-2^T and PR-3^T were closely related to previously known magnetotactic strains, *Magnetovibrio blakemorei* MV-1^T and *Magnetococcus marinus* MC-1^T, with 16S rRNA gene sequence similarity of 94.40% and 97.39%, respectively. Strain PR-1^T is a spirillum that produces two single flagella at each pole; strain PR-2^T is a vibrio equipped with a single polar flagellum, while strain PR-3^T is roughly spherical, with two sheathed bundles of flagella at a single pole (bilophotrichous). The major cellular fatty acids are C_{18:1} ω7c and C_{16:1} ω7c for strain PR-1^T; C_{18:1} ω7c, C_{16:1} ω7c and C_{16:0} for strain PR-2^T; and C_{16:1} ω7c and C_{16:0} for strain PR-3^T. Based on phenotypic, genotypic and phylogenetic data, all three strains, PR-1^T, PR-2^T and PR-3^T, are considered to represent three novel species. Strain PR-1^T belongs to the genus *Terasakiella*, for which the name *Terasakiella magnetica* sp. nov. is proposed. The type strain is PR-1^T (=TSD-367^T=JCM 39464^T). Strain PR-2^T belongs to the genus *Magnetovibrio*, for which the name *Magnetovibrio massiliensis* sp. nov. is proposed. The type strain is PR-2^T (=TSD-368^T=JCM 39465^T). Strain PR-3^T belongs to the genus *Magnetococcus*, for which the name *Magnetococcus organivorans* sp. nov. is proposed. The type strain is PR-3^T (=TSD-369^T=JCM 39466^T).

INTRODUCTION

Magnetotactic bacteria (MTB) are widespread and diverse bacteria capable of magnetotactic behaviour, referred to as magnetochemotaxis, due to the presence of biomineralized intracellular magnetic crystals [magnetite (Fe₃O₄) or greigite (Fe₃S₄)] surrounded by a biomembrane – magnetosomes [1]. MTB are microaerophilic micro-organisms usually found at the oxic–anoxic interface (OAI) in aquatic environments. It is assumed that magnetotaxis is a biological adaptation helping bacteria to rapidly find the microaerobic zone [2, 3]. MTB are affiliated with several phyla, such as *Pseudomonadota*, *Desulfobacterota*, *Nitrospirata*, *Omnitrophota*, *Latescibacterota*, *Hydrogenedentota* and *Elusimicrobiota* [4–6]. Due to strict growth requirements, few MTB species have been isolated in axenic culture, all of which belong to the *Pseudomonadota* and *Desulfobacterota* phyla. Thus, only a few species have been described and validly published. In the *Desulfobacterota*, validly published species are classified under

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Keywords: *Magnetococcus*; magnetotactic bacteria; *Magnetovibrio*; *Terasakiella*.

Abbreviations: AAI, average amino acid identity; AL, aminolipid; ANI, average nucleotide identity; dDDH, digital DNA–DNA hybridization; MLH, maximum-likelihood; MP, maximum-parsimony; MTB, magnetotactic bacteria; NJ, neighbour-joining; OAI, oxic–anoxic; PE, phosphatidylethanolamine; PG, phosphatidylglycerol; POCp, percentages of conserved proteins.

The GenBank accession numbers for the 16S rRNA gene sequence and whole-genome sequence of strain PR-1^T are KJ442651 and NZ_FLYE000000000, respectively. The GenBank accession numbers for the 16S rRNA gene sequence and whole-genome sequence of strain PR-2^T are KJ442652 and JBAHUR000000000, respectively. The GenBank accession numbers for the 16S rRNA gene sequence and whole-genome sequence of strain PR-3^T are KJ442653 and JBAHUQ000000000, respectively. American Type Culture Collection (ATCC) deposit numbers for strains PR-1^T, PR-2^T and PR-3^T are TSD-367, TSD-368 and TSD-369, respectively. Japan Collection of Microorganisms (JCM) deposit numbers for strains PR-1^T, PR-2^T and PR-3^T are JCM 39464, JCM 39465 and JCM 39466, respectively.

Two supplementary figures are available with the online version of this article.

Solidesulfovibrio (former *Desulfovibrio*; *Solidesulfovibrio magneticus* strain RS-1 [7]) and *Fundidesulfovibrio* (*Fundidesulfovibrio magnetotacticus* strain FSS-1 [8]) genera, while in *Pseudomonadota*, they fall within the *Magnetospirillum* (*Magnetospirillum gryphiswaldense* strain MSR-1 [9]), *Paramagnetospirillum* (*Paramagnetospirillum caucaseum* strain SO-1 [10]), *Magnetospira* (*Magnetospira thiophila* strain MMS-1 [11]), *Magnetovibrio* (*Magnetovibrio blakemorei* strain MV-1 [12]) and *Magnetococcus* (*Magnetococcus marinus* strain MC-1 [13]) genera, with most belonging to the first two genera [14]. In 2014, a study reported the isolation of three novel MTB strains – PR-1, PR-2 and PR-3 – from the Mediterranean Sea, which have not been described. PR-1 belonged to the genus *Terasakiella*, where MTB had not been previously found, while PR-2 and PR-3 were identified within *Magnetovibrio* and *Magnetococcus* genera, both of which already contained MTB representatives [3].

The genus *Terasakiella* of the family *Terasakiellaceae*, part of the order *Rhodospirillales*, was first described by Satomi *et al.* [15, 16]. The genus currently contains three species, *Terasakiella pusilla* NBRC 13613^T [15], *Terasakiella brassicae* KCTC 42652^T [17] and *Terasakiella salincola* KMU-80^T [18], which were isolated from a putrid infusion of marine shellfish, wastewater collected from a pickle-processing factory and seawater, respectively. These three species are S-shaped, halophilic, chemo-heterotrophic and polyhydroxybutyrate-accumulating bacteria. *T. pusilla* and *T. brassicae* are aerobic, while *T. salincola* is facultatively anaerobic. All three strains are motile by means of two single flagella at each pole, but magnetotaxis has not been reported. Two microaerophilic strains of the *Terasakiella* genus, which showed magnetotactic behaviour, *Candidatus Terasakiella magnetica* PR-1 [3, 19] and *Terasakiella* sp. SH-1 [20], were isolated from seawater.

The genus *Magnetovibrio* was first proposed by Bazylinski *et al.* and included one species, *M. blakemorei* MV-1^T, which was isolated from sulphide-rich sediments in a salt marsh [12, 21]. Strain MV-1^T is facultatively microaerophilic and anaerobic, capable of chemolithoautotrophic and chemo-organoheterotrophic growth. Cells are motile by means of a single polar flagellum and exhibit magnetotaxis [12].

Magnetotactic cocci of the class *Magnetococcia* [22, 23] generally represent the most abundant morphotype of MTB observed in the environment [24]. However, only one pure culture of freshwater magnetotactic cocci, *Ca. Magnetaquiglobus chichijimensis* strain FCR-1, was obtained, while most were described by a culture-independent approach, like *Ca. Magnetaquicoccus inordinatus* UR-1 [25, 26]. Several pure cultures of marine magnetotactic coccus were isolated and are affiliated with the genera *Magnetococcus* and *Ca. Magnetofaba* [13, 27, 28]. The genus *Magnetococcus* was first introduced by Bazylinski *et al.* [13] and comprised a single validly described species, *M. marinus* MC-1^T, isolated from the OAI of the Pettaquamscutt Estuary [13, 29]. Besides, the genus contains cultured but not validly described species *Ca. Magnetococcus massiliensis* MO-1 [30] and strain SS-1 [3], as well as numerous uncultured species, like *Ca. Magnetococcus yuandaduensis* [31]. Representatives of the genus are spherical/coccoid, microaerophilic and chemolithoautotrophic bacteria which are motile by means of two bundles of several flagella and show magnetotaxis.

In this study, we describe three strains of MTB, PR-1^T, PR-2^T and PR-3^T, isolated from the Mediterranean Sea. Based on the results of a comprehensive polyphasic approach, we propose that strains PR-1^T, PR-2^T and PR-3^T represent three novel species of the genera *Terasakiella*, *Magnetovibrio* and *Magnetococcus*, respectively.

SAMPLING SITE AND ISOLATION

Strains PR-1^T, PR-2^T and PR-3^T were isolated in 2011 during a single field trip seeking new MTB in the Mediterranean Sea. Mud and water samples collected from the Pointe Rouge Marina in Marseille, France (43.24555 N, 5.36603 E), contained significant MTB populations (>10³ cells ml⁻¹) of various morphologies as reported in previous studies [32, 33]. The majority of MTB were magnetotactic cocci, likely related to strain MO-1 isolated from the same location [33]. Several dilutions of magnetically purified cells were used as inoculum in a growth medium modified from Lefèvre *et al.* [33] to target microaerophilic and chemo-organoheterotrophic MTB. After several weeks of incubation at 25 °C, bands of cells were observed at the OAI of several tubes. These cells were helical to vibrioid or coccoid to ovoid in morphology and displayed a magnetotactic behaviour under the microscope. Axenic cultures of PR-1^T, PR-2^T and PR-3^T were obtained after diluting a cell band to extinction, and culture axeny was verified several times by sequencing the 16S rRNA gene in multiple clones. Cells were routinely grown in the same semisolid oxygen-gradient medium (used as a positive control for the growth tests). The formula of the artificial seawater used in this medium was (per litre): 20.0 g NaCl, 6.0 g MgCl₂•6H₂O, 3.24 g Na₂SO₄, 1.0 g CaCl₂•2H₂O and 0.5 g KCl. To a litre of this mixture, 5 ml modified Wolfe's mineral elixir [34, 35], 50 µl 1% aqueous resazurin, 0.3 g NH₄Cl, 1 g sodium succinate, 1.26 g of NaHCO₃, 2.4 g of HEPES and 0.2 g yeast extract were added, and the pH was adjusted to 7.0 with 1 M HCl. 2.0 g of Bacto-Agar (Difco Laboratories, Detroit, MI) was then added after which the medium was autoclaved. After autoclaving, 0.5 ml of a stock solution of vitamins [36], 1.8 ml of 0.5 M KHPO₄ buffer (pH 7.0), 3 ml of 10 mM FeCl₂•4H₂O (in 0.02 M HCl) and 0.4 g of freshly made filtered sterilized cysteine were added to the medium. Screw-capped glass culture tubes were filled to ~80% of their volume with medium. The medium was allowed to sit at room temperature for several hours to solidify and to allow the O₂ concentration gradient to form as evidenced by the presence of a pink (oxidized) zone near the surface and a colourless (reduced) zone in the lower part of the tubes, thanks to the redox indicator resazurin. The medium was generally inoculated with 50 µl of a band of cells from a culture in the exponential growth phase and incubated at 28 °C.

16S rRNA PHYLOGENETIC ANALYSIS

First, the identification of PR-1^T, PR-2^T and PR-3^T was performed by aligning 16S rRNA gene sequences against a public database using BLASTN algorithm. Sequences were obtained after PCR amplification using bacteria-specific primers 27F 5-AGAGTTT-GATCMTGGCTCAG-3 and 1492R 5-TACGGHTACCTTGTTACGACTT-3 [37]. For checking culture axeny, blunt-end PCR products were cloned using a Zero Blunt[®] TOPO[®] PCR Cloning Kit with One Shot[®] TOP10 chemically competent *E. coli* cells. Five clones were sequenced to validate the clonal population of the three strains.

The 16S rRNA gene sequence comparison of strains PR-1^T, PR-2^T and PR-3^T was conducted using EzBioCloud [38]. Strain PR-1^T showed 97.9%, 96.8%, 96.1% and 94.9% sequence similarities with the members of the genus *Terasakiella* – *T. brassicae* B3^T, *Terasakiella* sp. SH-1, *T. pusilla* DSM 6293^T and *T. salincola* KMU-80^T, respectively. Strain PR-2^T shared 96.0% 16S rRNA sequence similarity with *M. blakemorei* MV-1^T, while strain PR-3^T showed 97.8% and 94.0% similarities with *M. marinus* MC-1^T and *Ca. Magnetococcus massiliensis* MO-1, respectively. The 16S rRNA gene sequences were aligned using MUSCLE [39]. The maximum-likelihood (MLH) tree of the 16S rRNA genes was constructed using IQ-TREE [40] with the GTR+F+I+G4 model, as recommended by ModelFinder [41]. Branch support was assessed with 10,000 ultrafast bootstraps [42]. Additionally, maximum-parsimony (MP) and neighbour-joining (NJ) trees were reconstructed using MPBoot [43] and MEGA11 [44], respectively.

On the phylogenetic tree (Fig. 1) constructed using the MLH, NJ and MP methods, strain PR-1^T formed a clade with *Terasakiella* species, PR-2^T clustered with *M. blakemorei* MV-1^T, and PR-3^T grouped with *Magnetococcus* species.

GENOMIC CHARACTERIZATION

For each strain, genomic DNA was extracted using the DNeasy Blood and Tissue Kit from QIAGEN on three bands of cells in the exponential growth phase grown in the heterotrophic semisolid oxygen concentration gradient medium. Genome sequencing (2×300 bp) was performed with a MiSeq Illumina sequencer by GATC Biotech (Konstanz, Germany) in 2015. Genome assembly was performed as previously mentioned [45] with the AMALGAM assembly pipeline (Automated Microbial Genome Assembler) [46]. Qualities of obtained assemblies were assessed using the Quast software [47]. CDSs were automatically and manually annotated using the MicroScope platform from the Genoscope [48]. The genome completeness was assessed using CheckM [49]. Assembly results for the three strains are shown in Table 1. Strain PR-1^T genome features are closely related to those of the *Terasakiella* genus (e.g. *T. pusilla* with a genome length of 4,061,960 bases with a GC% of 50.02). Strain PR-2^T genome features are similar to those of *M. blakemorei* MV-1^T (e.g. genome length of 3,619,470 bases with a GC% of 54.42). The genome size of strain PR-3^T is slightly larger compared to those of the members of the *Magnetococcus* genus (e.g. *M. marinus*: 4,719,581 bases) and *Magnetofaba australis* (i.e. 4,988,701 bases). While the GC% of strain PR-3^T is slightly smaller compared to those of the members of the *Magnetococcus* genus (e.g. *M. marinus*: 54.17%) and *M. australis* (i.e. 61.30%).

Phylogenomic analysis of the strains was conducted using a concatenated alignment of 120 single-copy phylogenetic marker genes obtained using the software GTDB-Tk version 1.4.0 [50]. The MLH phylogenomic tree was calculated using IQ-Tree [40] with the model recommended by ModelFinder, and branch support was estimated using UFBoot2 [42]. MP and NJ trees were reconstructed using MPBoot [43] and MEGA11 [44], respectively. The pair-wise average nucleotide identity (ANI) and *in silico* digital DNA–DNA hybridization (dDDH) values among PR strains and closely related type strains were calculated based on their whole genomes using FastANI [51] and Genome-to-Genome Distance Calculator version 2.1 [52], with BLAST+ for genome alignments [53], respectively. Average amino acid identity (AAI) values were calculated by CompareM 0.0.23 (<https://github.com/dparks1134/CompareM>) with default BLASTP parameters (i.e. e-value ≤0.001, per cent identity ≥30% and alignment length ≥70%). Pairwise percentages of conserved proteins (POCP) were calculated using the runPOCP.sh script [54], based on a previously described approach [55].

On the phylogenomic tree, strain PR-1^T formed a branch with *Terasakiella* sp. SH-1, *T. pusilla* DSM 6293^T and *T. brassicae* CGMCC 1.15254^T (Fig. 2). The ANI and dDDH values compared with the validly described *Terasakiella* species did not exceed 78.6% and 19.4% (Table 2), respectively, and were below the species cutoffs of 95–96% for ANI and 70% for dDDH [56]. Notably, the magnetotactic strain SH-1 also differs at the species level from strain PR-1^T. The AAI (75.8–82.0%) and POCP (66.5–77.1%) values for *Terasakiella* spp. strain PR-1^T exceed the genus delineation thresholds of 71% and 50–60%, respectively [14, 55]. Based on the findings mentioned above, we propose that strain PR-1^T represents a novel species of the genus *Terasakiella*.

Similarly, the genomic indices (Table 2) and phylogenetic analyses (Figs 1 and 2) strongly indicate that strains PR-2^T and PR-3^T represent new species within the genera *Magnetovibrio* and *Magnetococcus*, respectively. The low ANI (<78.3%) and dDDH (<24.1%) values relative to their closest known species, combined with phylogenetic placement, reinforce their classification as distinct species. Meanwhile, the AAI (>72.6%) and POCP (67.4%) values confirm that these strains belong to their respective genus, further supporting their taxonomic assignment.

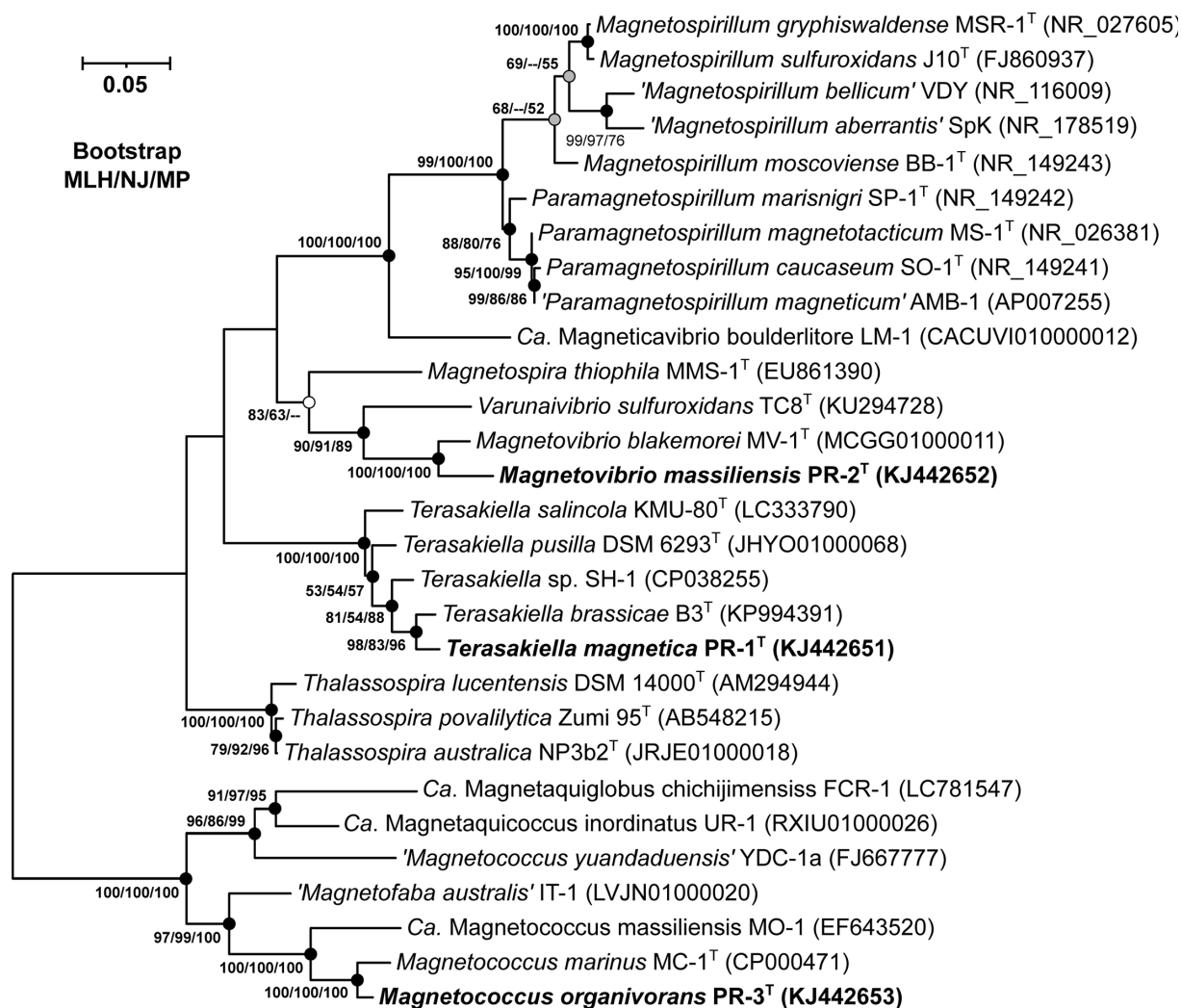


Fig. 1. MLH phylogenetic tree based on 16S rRNA gene sequences (1,441 nucleotide sites) reconstructed with evolutionary model GTR+F+I+G4, showing the position of strains PR-1^T, PR-2^T and PR-3^T. Grey circles indicate that the corresponding nodes were recovered in the reconstructed tree based on the MP algorithm; white circles indicate that the corresponding nodes were recovered using the NJ algorithm; black circles indicate that the corresponding nodes were also recovered based on the NJ and MP algorithms. Bootstrap values (>50%) are listed as percentages at the branching points. Bar, 0.05 substitutions per nucleotide position. GenBank accession numbers for 16S rRNA genes are indicated in brackets. The name of the strains obtained in this study is marked in bold.

PHYSIOLOGICAL AND CHEMOTAXONOMIC CHARACTERIZATION

Cells of strain PR-1^T are spiral-shaped with a single or double helix turn. The ultrastructure and motility of this strain have been previously described [45]. PR-1^T biomineralizes a single chain of elongated prismatic crystals of magnetite in their magnetosomes, evenly spaced from each other and positioned along the long axis of the cell (Fig. 3a–c). Magnetosome features of strain PR-1^T resembled those described for *M. blakemorei* strain MV-1^T [12]. Cells of strain PR-1^T are motile and magnetotactic and possess a single polar flagellum at each pole of the cell (Fig. 3a). Strain PR-1^T presents a magnetotactic behaviour that combines unipolar magneto-aerotaxis directed toward the oxic zone and axial magnetotaxis [3]. Although it possesses two single flagella at each pole, motility, as well as magnetotaxis features of strain PR-1^T, is similar to those described for *M. blakemorei* strain MV-1^T [3]. Strain PR-2^T is very similar to *M. blakemorei* considering the ultrastructure, cell motility and magnetotactic behaviour [3, 12]. Cells of strain PR-2^T are small vibrios with a single polar flagellum, synthesizing a single chain of prismatic elongated spaced magnetosomes (Fig. 3d–f). Considering the ultrastructure, cell motility and magnetotactic behaviour, strain PR-3^T is very similar to previously cultured magnetotactic cocci *M. marinus* strain MC-1^T [13], *Ca. Magnetococcus massiliensis* strain MO-1 [33, 57] and *Ca. Magnetofaba australis* strain IT-1 [28]. Cells have a coccoid to ovoid morphology with two bundles of seven flagella at a single pole (bilophotrichous) (Fig. 3g). A single chain of prismatic magnetite magnetosomes is produced (Fig. 3g–i). Cells are fast motile using a unique helical path as described in several biophysical studies (e.g. [58]). Ultrathin sections of a pellet of cells

Table 2. Genome relatedness indexes (%) between strains PR-1^T, PR-2^T, PR-3^T and closely related bacteria

Genome	Reference genome	ANI	dDDH	AAI	POCP
PR1 ^T	<i>Terasakiella</i> sp. SH-1	79.9	20.3	82.0	77.1
	<i>T. brassicae</i> CGMCC 1.15254 ^T	78.4	18.7	77.8	71.3
	<i>Terasakiella pusilla</i> DSM 6293 ^T	78.6	19.4	75.8	66.5
PR2 ^T	<i>M. blakemorei</i> MV-1 ^T	78.3	18.6	72.6	67.4
	<i>Varunaivibrio sulfuroxidans</i> DSM 101688 ^T	<70	20.0	61.9	55.9
	<i>Magnetospira</i> sp. QH-2	<70	20.5	59.1	48.4
PR3 ^T	<i>M. marinus</i> MC-1 ^T	78.2	19.2	76.3	69.0
	<i>Ca. Magnetococcus massiliensis</i> MO-1	<70	24.1	66.9	65.1
	<i>Ca. Magnetofaba australis</i> strain IT-1	<70	21.8	60.2	51.2
	<i>Ca. Magnetaquicoccus inordinatus</i> UR-1	<70	20.5	51.2	34.2
	<i>Ca. Magnetaquiglobus chichijimensis</i> FCR-1	<70	26.1	51.0	39.9

For fatty acid, respiratory quinone and polar lipid analyses, several bands of cultures in the exponential growth phase of the three strains were harvested and centrifuged at 16,500 *g* during 5 min at 20 °C. The pellets were conserved at –80 °C and were then lyophilized. Fatty acid, respiratory quinone and polar lipid analyses were carried out by the Identification Service of the Deutsche Sammlung von Mikroorganismen und Zellkulturen GmbH (DSMZ), Braunschweig, Germany. Fatty acids were analysed after conversion into fatty acid methyl esters by saponification, methylation and extraction [59]; respiratory quinones were analysed by HPLC using a reverse-phase column recording absorption spectra [60]; and polar lipids were detected using spray reagents specific for defined functional groups [61, 62].

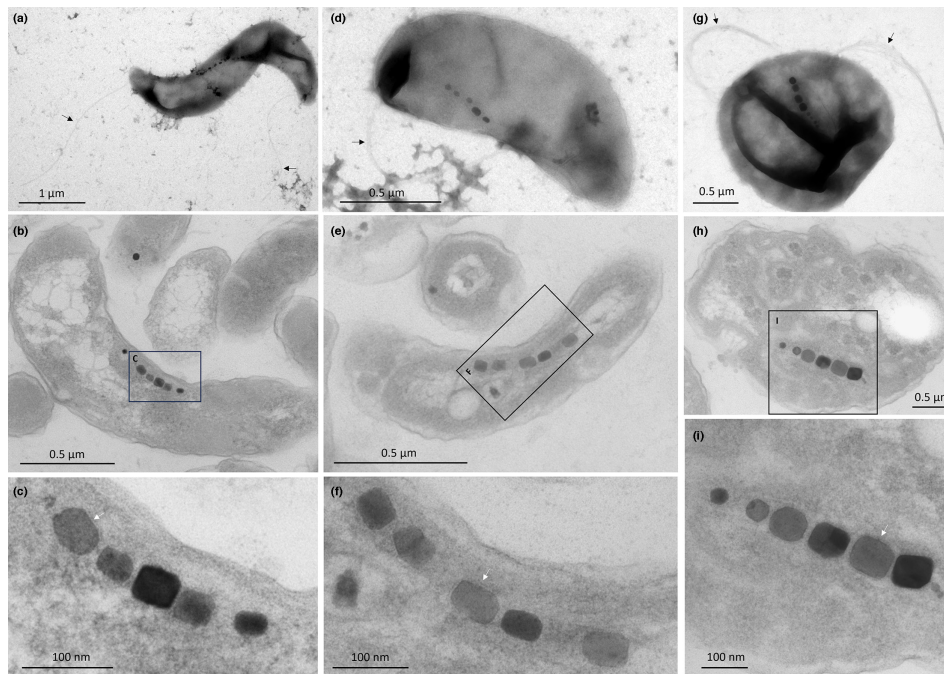


Fig. 3. Transmission electron microscopy images of cells of strains PR-1^T, PR-2^T and PR-3^T. Whole cells stained with 1% uranyl acetate showing the flagellar apparatus shown at the black arrow of strains PR-1^T (a), PR-2^T (d) and PR-3^T (g). Ultrathin sections of cells of strains PR-1^T (b and c), PR-2^T (e and f) and PR-3^T (h and i) showing a membrane surrounding the magnetite particles (white arrows). The frames in images b, e and h correspond to the higher magnified images c, f and i, respectively.

Catalase activity was assessed qualitatively by monitoring the formation of oxygen bubbles upon addition of cell suspensions to a 3% (v/v) aqueous hydrogen peroxide solution. Oxidase activity was evaluated using N,N,N',N'-tetramethyl-p-phenylenediamine dihydrochloride-impregnated oxidase test discs (Sigma-Aldrich), following the manufacturer's instructions. *Escherichia coli* was used as a positive control.

All three strains were slight halophiles that preferred 2–4% of NaCl in growth media. PR strains grew within a temperature range of 20–30 °C, with an optimum at 28 °C, whereas their closely related strains exhibited a broader growth temperature range. In addition, the optimum growth temperature of strain B3^T was shifted towards higher, more thermophilic conditions, compared to PR-1^T. The PR strains differed from their closest relatives in substrate utilization. The growth substrate spectrum of PR-1^T was narrower compared to B3^T. Strain PR-2^T showed a similar pattern to MV-1^T, although MV-1^T could utilize pyruvate and PR-2^T showed weak growth on fructose. Strain PR-3^T grew chemo-heterotrophically with succinate, unlike its closely related strain MC-1^T which grew on acetate and could use thiosulphate and sulphide as electron donors. Detailed results of physiological and biochemical tests are given in the species description. The differential characteristics between PR strains and the reference strains are summarized in Table 3.

All PR strains contained monounsaturated fatty acids as major components; however, their dominant profiles showed a distinct pattern compared to their reference strains. Strain PR-1^T contained mainly oleic (C_{18:1} ω7c) and palmitoleic (C_{16:1} ω7c) acids, whereas B3^T showed a more complex profile dominated by palmitoleic acid together with 2-hydroxy iso-pentadecanoic acid (iso-C_{15:0} 2-OH). PR-2^T included oleic, palmitoleic and palmitic (C_{16:0}) acids, while MV-1^T additionally contained *cis*-vaccenic acid (C_{18:1} ω5). Strains PR-3^T and MC-1^T shared the same major acids – palmitoleic, palmitic and oleic acids – but differed in their relative abundances. The polar lipid analysis of cell membranes of strains PR-1^T and PR-2^T showed the presence of two major polar lipids, phosphatidylethanolamine (PE) and phosphatidylglycerol (PG) and minor uncharacterized lipids and aminolipid (AL). Strain B3^T exhibited a complex polar lipid profile consisting of PE, PG, diphosphatidylglycerol, phosphatidylserine and multiple unidentified lipids and phospholipids [17]. Strain PR-3^T was characterized primarily by an uncharacterized AL and a minor uncharacterized lipid. Thin-layer chromatograms showing total polar lipid composition of PR strains are shown in Fig. S1, available in the online Supplementary Material. The respiratory quinone analysis showed ubiquinone-10 and ubiquinone-9 for PR-1^T, whereas B3^T contained only Q-10. PR-2^T carried Q-9, Q-10 and Q-8. Strains PR-3^T and MC-1^T both contained the same single quinone Q-8.

All PR strains, as well as B3^T, MV-1^T and MC-1^T, were oxidase-positive, indicating cytochrome *c* oxidase activity. In contrast, catalase activity differentiated the PR strains from their closely related strains. Strains PR-1^T and PR-2^T were catalase-positive, while B3^T and MV-1^T were catalase-negative. Both PR-3^T and MC-1^T showed negative catalase reactions. Results of oxidase and catalase tests are shown in Fig. S2.

In conclusion, the three magnetotactic bacterial isolates from seawater are clearly different from their nearest phylogenetic relatives and are proposed to be classified as representatives of the novel species: *Terasakiella magnetica* sp. nov., *Magnetovibrio massiliensis* sp. nov. and *Magnetococcus organivorans* sp. nov.

SPECIES DESCRIPTION

Emended description of the genus *Terasakiella* Satomi et al. 2002

The description is as given before [15, 17, 18] with additions. Some species show magneto-aerotactic behaviour. Species are aerobic, microaerophilic or facultatively anaerobic.

Description of *Terasakiella magnetica* sp. nov.

Terasakiella magnetica (mag.ne'ti.ca. L. fem. adj. magnetica, magnetic).

Cells are Gram-negative, spiral-shaped with a single or double helix turn, contain a single chain of elongated prismatic crystals of magnetite, are motile and magnetotactic and possess a single flagellum at each pole. Cells are microaerophilic and slightly halophilic, requiring 2–4% NaCl for growth. The optimum temperature for growth is 28 °C, and the pH growth range is 7.0–8.0. Chemo-organoheterotrophic growth occurs under microaerobic conditions with succinate and malate. Formate, acetate, tartrate, lactate, oxalate, pyruvate, glucose, fructose, maltose, sucrose, trehalose and fucose are not utilized. Oxygen is used as a terminal electron acceptor; nitrite, nitrate and N₂O are not utilized. Cellular fatty acids are dominated by C_{18:1} ω7c and C_{16:1} ω7c. The polar lipids are dominated by PE and PG. The respiratory quinones are Q-10 and Q-9. The strain is oxidase- and catalase-positive. The genome consists of a single, circular chromosome with a size of 3.7 Mb and a G+C content of 46.0 mol%.

The type strain, PR-1^T (=TSD-367^T=JCM 39464^T), was isolated from mud and water samples collected from the Pointe Rouge Marina in Marseille, France. The GenBank accession number for the 16S rRNA gene sequence is KJ442651. The GenBank accession number for the whole-genome sequence is NZ_FLYE00000000.

Table 3. Comparative properties of the PR strains and their closest relatives: *T. brassicae* B3^T, *M. blakemorei* MV-1^T and *M. marinus* MC-1^T, respectively

Property	PR-1 ^T	PR-2 ^T	PR-3 ^T	B3 ^T	MV-1 ^T ‡	MC-1 ^T §
Cell morphology	S-shaped	Small vibrio	Spherical/coccoid	S-shaped, helical (1.5–4×0.3–0.5 µm)	Vibrioid-helical (1–3×0.2–0.4 µm)	Spherical/coccoid (1–2 µm in diameter)
Formation of magnetosomes	+	+	+	–	+	+
Relationship to oxygen	Microaerophilic	Microaerophilic	Microaerophilic	Aerobic	Facultatively microaerophilic and anaerobic	Obligate microaerophilic
NaCl concentration range, w/v	2–4%	2–4%	2–3%	0.5–8%	1–3%	1–3%
pH range (optimum)	7.0–8.0 (7.0)	6.5–8.0 (7.0)	7.0–7.5 (7.0)	5.5–9.0 (6.5–7.5)	5.5–7.5 (7.0)	6.0–7.5 (7.0)
Growth temp. range (optimum), °C	20–28 (28)	20–28 (28)	20–28 (28)	4–40 (32–37)	4–31 (26–28)	5–35 (25)
Electron donors	Organic C	Thiosulphate, organic C	Organic C	Organic C	Thiosulphate, organic C	Thiosulphate, sulphide, organic C
Growth substrates	Succinate, malate	Succinate, malate, formate, acetate, tartrate, lactate, fructose*	Succinate	Succinate, malate, acetate, pyruvate, lactate, propionate, butyrate	Succinate, malate, formate, acetate, tartrate, lactate, pyruvate	Acetate
Acceptors	O ₂	O ₂	O ₂	O ₂	O ₂ , nitrate, nitrite, N ₂ O	O ₂
Dominant fatty acids (in order of dominance)	C _{18:1} ω7c, C _{16:1} ω7c	C _{18:1} ω7c, C _{16:1} ω7c, C _{16:0}	C _{16:1} ω7c, C _{16:0}	C _{16:1} ω7c/iso-C _{15:0} 2-OH, C _{18:1} ω7c, C _{16:0}	C _{18:1} ω7c, C _{16:1} ω7c, C _{16:0} ω5	C _{16:1} ω7c, C _{16:0} ω7c, C _{18:1} ω7c
Predominant quinones	Q-10, Q-9	Q-9, Q-10, Q-8	Q-8	Q-10	ND	Q-8
Major polar lipids	PE, PG	PE, PG	AL	DPG, PG, PE	ND	ND
Catalase activity	+	+	–	–	–	–
Oxidase activity	+	+	+	+	+	+
DNA G+C content (mol%)	46.0	54.0	50.6	42.3	53.5	52.9–53.5

*week growth.

†Data from Han et al. [17].

‡Data for cellular fatty acids, quinones and polar lipids were taken from Bazylynski et al. [12].

§Data for cellular fatty acids and polar lipids were taken from Bazylynski et al. [13], while data for quinones were taken from Chobert et al. [63].

AL, aminolipid; DPG, diphosphatidylglycerol; ND, not determined; PE, phosphatidylethanolamine; PG, phosphatidylglycerol.

Description of *Magnetovibrio massiliensis* sp. nov.

Magnetovibrio massiliensis (mas.si.li.en'sis. L. masc. adj. *massiliensis*, pertaining to Marseille).

Cells are small vibrios with a single polar flagellum and synthesize a single chain of prismatic elongated spaced magnetosomes. Cells are microaerophilic and slightly halophilic, requiring 2–4% NaCl for growth. The optimum temperature for growth is 28 °C, and the pH growth range is 6.5–8.0. The strain is capable of autotrophic growth with thiosulphate as electron donor. Chemo-organoheterotrophic growth occurs under microaerobic conditions with succinate, malate, formate, acetate, tartrate, lactate and fructose. Oxalate, pyruvate, glucose, maltose, sucrose, trehalose and fucose are not utilized. Oxygen is used as a terminal electron acceptor; nitrite, nitrate and N₂O are not utilized. Cellular fatty acids are dominated by C_{18:1} ω7c, C_{16:1} ω7c and C_{16:0}. The polar lipids are dominated by PE and PG. The respiratory quinones are Q-10, Q-9 and Q-8. The strain is oxidase- and catalase-positive. The genome consists of a single, circular chromosome with a size of 3.7 Mb and a G+C content of 54.0 mol%.

The type strain, PR-2^T (=TSD-368^T=JCM 39465^T), was isolated from mud and water samples collected from the Pointe Rouge Marina in Marseille, France. The GenBank accession number for the 16S rRNA gene sequence is KJ442652. The GenBank accession number for the whole-genome sequence is JBAHUR000000000.

Emended description of the genus *Magnetococcus* Bazylnski et al. 2013

The description is as given before [13] with additions. Organic C is used as energy source during heterotrophic growth.

Description of *Magnetococcus organivorans* sp. nov.

Magnetococcus organivorans (or.ga.ni.vo'rans. L. neut. n. *organum*, organic compound; L. v. *vorare*, to eat, to consume; N.L. masc. part. adj. *organivorans*, devouring organic compounds).

Cells have a coccoid to ovoid morphology with two bundles of seven flagella at a single pole, producing a single chain of prismatic magnetite magnetosomes. Cells are microaerophilic, slight halophilic, which requires 2–3% NaCl for growth. The optimum temperature for growth is 28 °C, and the pH growth range is 7.0–7.5. Chemo-organoheterotrophic growth occurs under microaerobic conditions using succinate. Malate, formate, acetate, tartrate, lactate, oxalate, pyruvate, glucose, fructose, maltose, sucrose, trehalose and fucose are not utilized. Oxygen is used as a terminal electron acceptor; nitrite, nitrate and N₂O are not utilized. Cellular fatty acids are dominated by C_{16:1} ω7c and C_{16:0}. The polar lipids are dominated by AL. The respiratory quinone is Q-8. The strain is oxidase-positive, and catalase-negative. The genome consists of a single, circular chromosome with a size of 5.5 Mb and a G+C content of 50.6 mol%.

The type strain, PR-3^T (=TSD-369^T=JCM 39466^T), was isolated from mud and water samples collected from Pointe Rouge Marina in Marseille, France. The GenBank accession number for the 16S rRNA gene sequence is KJ442653. The GenBank accession number for the whole-genome sequence is JBAHUQ000000000.

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Conflicts of interest

The authors declare that there are no conflicts of interest.

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